

VARUNA (SIH26143) — Plain-English & Easy Technical Guide

Vessel Attribution through Remote-sensing & Unified Navigational Analytics

Smart India Hackathon 2026 — Problem Statement **SIH26143**

1. What is VARUNA? (The Ocean Detective Story)

Imagine a ship secretly dumps oil in the ocean at night and sails away. How do authorities catch who did it?

VARUNA works like an **AI Maritime Detective**:

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A[" Step 1: Take a Satellite Photo"] --> B[" Step 2: Find Dark Oil Stains"]

B --> C[" Step 3: Human Analyst Checks the Stain"]

C --> D[" Step 4: Rewind Time (Simulate Wind & Ocean Currents)"]

D --> E[" Step 5: Mark the Release Zone & Time Window"]

E --> F[" Step 6: Match GPS Trajectories of Ships Nearby"]

F --> G[" Step 7: Print an Evidence Report with Suspect Ranks"]



The Satellite Photo: A radar satellite takes a photo of the ocean. Oil smooths out waves, so oil spills look like dark black spots.

Spotting the Spill: The system scans the image and highlights dark spots.

Rewinding Time (Drift Physics): Oil floats on water. We drop 5,000 virtual particles on the oil spill and **rewind the clock backward in time** using real ocean current and wind data to see where the oil came from.

Checking Ship GPS (AIS Tracks): We look at all ships that passed through that exact spot at that exact time.

Evaluating 12 Clues: We grade each ship on 12 clues (e.g., *How close was it? Did it turn off its GPS transmitter? Did its ship weight change?*).

6. The Final Report: The system ranks the suspect ships and prints a complete, audit-proof evidence report.

 2. How Was the Project Built? (SDLC Model)

VARUNA was built using a **Phase-Gated Agile MVP (Minimum Viable Product) Model**.

What does this mean? Instead of building everything at once, the project was built step-by-step in **14 distinct phases** (Phase 0 to Phase 13).

The Rule: Phase 1 (Geodesy distance calculators and Data schemas) had to be 100% tested and verified before starting Phase 2 (User Login & Job Queues), and so on.

 3. Algorithms Explained in Plain English

Here are the main algorithms used in VARUNA and why each one was chosen:

Stage	Algorithm Name	Plain English Explanation	Why This Algorithm?
Spotting Oil Slicks	Darkspot-v1 (Local Adaptive Thresholding)	Compares each pixel on the satellite photo against its local ocean background to find spots that are at least 3 dB darker than the surrounding sea.	A single global threshold fails because satellite photos get darker near the edges. Comparing local neighborhoods works everywhere on the image.
Calculating Distances	Karney's WGS84 Ellipsoidal Geodesy	Calculates distances and areas on the real curved, oval-shaped Earth (ellipsoid) rather than a flat map or perfect sphere.	Simple flat-map formulas can be 0.5% wrong over long distances. Karney's math guarantees accuracy under 0.1%.
Rewinding Time	Stochastic 2D Reverse Lagrangian Particle Integrator	Drops 5,000 virtual particles on the slick and pushes them backward in time (-15 minute steps) using ocean current vectors and wind speed.	Fixing wind drift at a single number gives a fake sense of precision. Randomly sampling wind and deflection angles gives an honest, realistic spill zone.
Spill Zone Mapping	Kernel Density Estimation (KDE)	Takes the scattered 5,000 particles and draws smooth 50% and 90% probability boundary circles around where they cluster.	Converts scattered dots into a clean, colored map layer showing where the leak most likely started.
Ship Scoring	12-Feature Weighted Scoring Engine	Grades candidate ships from 0 to 100 based on 12 weighted clues.	Renormalizes scores over *measured clues only*. If public data is missing a ship's type, the ship isn't unfairly punished for missing data.
Confidence Bounds	Monte Carlo Bootstrap Resampling	Runs 300 random simulations of the data to calculate error margins (e.g. 85% ± 4%).	Tells investigators whether Ship #1 is *definitely* guilty or if Ship #1 and Ship #2 are too close to call.

4. Model Accuracy & Evaluation Results

The `darkspot-v1` detector was tested on **66 held-out test satellite scenes** (22 oil spills, 22 look-alikes like calm wind zones, 22 clean seas):

Oil Detection Rate: 100% (It detected all 22 oil spills; 0 were missed).

Mean Oil IoU (Overlap Accuracy): 0.56 (Median: **0.63**).

Look-alike False Positives: 68% (It triggered on 15 out of 22 false look-alike features, such as low-wind calm sea zones).

 **Why Human Review is Mandatory:** Because low-wind ocean zones look dark like oil on radar, the detector finds all dark spots but relies on a human analyst to review and confirm `CONFIRM` , `EDIT` , or `REJECT` before taking action.

5. Datasets & How Bias Was Prevented

Main Dataset: Trujillo-Acatitla et al. Part III dataset (450 dual-polarization Sentinel-1 radar scenes).

Geographic Splitting: The data was divided by **ocean grid regions** (315 training scenes, 69 validation scenes, 66 test scenes). The test scenes came from geographic areas the model **never** saw during training, preventing regional bias.

Real Historical AIS Data: Uses real NOAA Marine Cadastre AIS data (US coastal waters) and Danish Maritime Authority (DMA) data. Zero fake/synthetic data is permitted.

6. Parameters & Formulas Made Simple

1. Look-Alike Risk Score

Scores whether a dark spot is real oil or just a calm water look-alike (0 = Oil, 1 = Look-alike):

Shape: Real oil slicks are stretched long by wind and currents; calm water spots are round blobs.

Edges: Real oil slicks have rough, jagged edges; look-alikes have smooth edges.

Contrast: Oil damps waves heavily, making the spot significantly darker than the sea around it.

2. Wind Suitability Gate

Wind < 2 m/s: The ocean is glassy calm everywhere. Everything looks like oil $\rightarrow$ **Confidence drops to 5%.**

Wind 4–9 m/s: Perfect wind for detection $\rightarrow$ **100% Confidence.**

Wind > 14 m/s: Waves re-rough the sea and wash away the slick $\rightarrow$ **Confidence drops to 5%.**

7. Time & Space Complexity (Speed & Memory)

Subsystem	Operation	Time Complexity (Speed)	Space Complexity (Memory)
SAR Segmentation	Local Median Filter & Threshold	$O(N)$ — Fast (~1 to 2 seconds for a full satellite image)	$O(N)$ — Stores image pixel array
Drift Integration	5,000 particles x 96 time steps	$O(P * T)$ — ~480,000 steps (~100 milliseconds)	$O(P * T)$ — Stores particle tracks
KDE Release Zone	Gaussian Kernel Density	$O(P * G)$ — Computes density grid	$O(G)$ — Stores grid map
Vessel Correlation	12 Clues + 300 Resample Runs	$O(M * V * K)$ — Fast execution for candidate vessels	$O(V * K)$ — Stores ship GPS coordinates

8. Zero Mock Data & Real-Data Policy

VARUNA has a **Strict Real-Data Policy**:

Zero fake, synthetic, or placeholder data is allowed anywhere in the system.

Every single satellite scene, AIS position fix, and current vector carries a **provenance record** (`sourceType` , `retrievedAt` , `externalId` , `checksum`).

If data is missing provenance, VARUNA's API automatically **strips the object** and flags it in the user interface.